Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

Make up

~~End Semester~~ Examination

2019–20

Max. Marks: 60 Duration: 180 Min

Class: B.E. Semester: VII

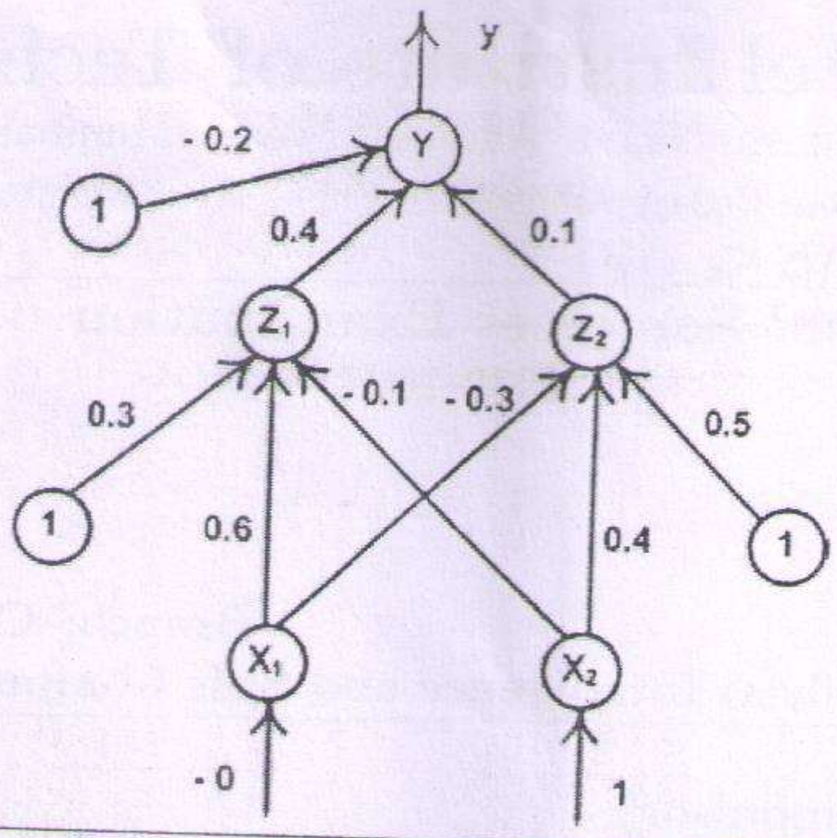
Course Code: CE71 Branch: Computer Engineering

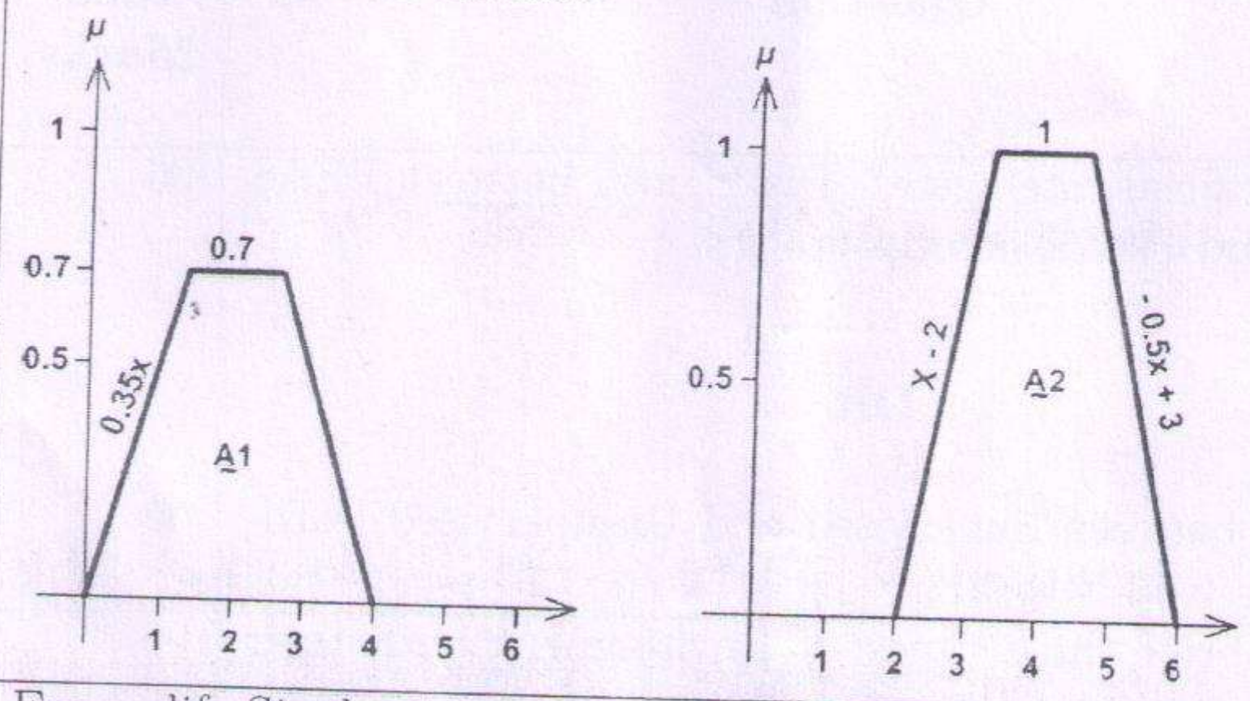
Name of the Course: Artificial Intelligence and Soft Computing

Instruction:

(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Q No.	Question	Max. Marks	CO-BL-PI
Q.1 (a)	What is Artificial Intelligent? Distinguish between PEAS descriptor and PAGE descriptor.	06	1-3-2.2.4
	OR		
	Distinguish between Supervised and Unsupervised learning.	06	1-3-2.2.4
Q.1 (b)	Explain in brief three types of Environment with their properties.	06	1-3-2.2.4
Q. 2	Consider an Learning Vector Quantization (LVQ) net with five vectors assigned to two classes. The given vector along with the class are as shown below: Value — Class [0 0 1 1] — 1 [1 0 0 0] — 2 [0 0 0 1] — 2 [1 1 0 0] — 1 [0 1 1 0] — 1 From the given five vectors, use the first two vectors as initial weight vectors and remaining vectors as input vectors. Construct LVQ net and Find the final weight matrix.	12	2-3-2.4.1
	OR		
	Using back-propagation network, find the new weights for the net shown in the following Figure. It is presented with the input pattern [0, 1] and the target output is 1. Use a learning rate $\alpha = 0.25$ and binary sigmoidal activation function.	12	2-3-2.4.1



Q.3 (a)	Define Fuzzy Relation and then exemplify it using Bipartite graph.	06	3-1-2.4.1
Q.3 (b)	Determine the defuzzified output value for the membership function shown below in figure using **any one method**: i) Center of largest area method ii) Center of sums method 	06	3-3-2.4.1
Q.4 (a)	Exemplify Single-site cross over and Multi-point cross over.	06	4-3-2.4.1
Q.4 (b)	Discuss the three phases of Genetic Algorithm namely i) Population Initialization, ii) Selection and iii) Reproduction to maximize the function $f(x) = x^2$ in the range of $x = 0$ to 5. [Note - Assume suitable data, if needed]	06	4-4-2.2.4
Q.5 (a)	Discuss Adaptive Neuro-Fuzzy Inference System.	06	5-4-2.2.4
Q.5 (b)	Using Zadeh's notation, determine the λ-cut sets for the given fuzzy sets: $\underset{\sim}{S}_1 = \left\{\frac{0}{0} + \frac{0.5}{20} + \frac{0.65}{40} + \frac{0.85}{60} + \frac{1.0}{80} + \frac{1.0}{100}\right\}$ $\underset{\sim}{S}_2 = \left\{\frac{0}{0} + \frac{0.45}{20} + \frac{0.6}{40} + \frac{0.8}{60} + \frac{0.95}{80} + \frac{1.0}{100}\right\}$ Express the following for $\lambda = 0.5$: i) $\left(\underset{\sim}{S}_1 \cup \underset{\sim}{S}_2\right)$; ii) $\left(\underset{\sim}{S}_1 \cap \underset{\sim}{S}_2\right)$; iii) $\left(\overline{\underset{\sim}{S}_1}\right)$; iv) $\left(\overline{\underset{\sim}{S}_2}\right)$; v) $\left(\overline{\underset{\sim}{S}_1 \cup \underset{\sim}{S}_2}\right)$; vi) $\left(\overline{\underset{\sim}{S}_1 \cap \underset{\sim}{S}_2}\right)$;	06	3-3-2.4.1